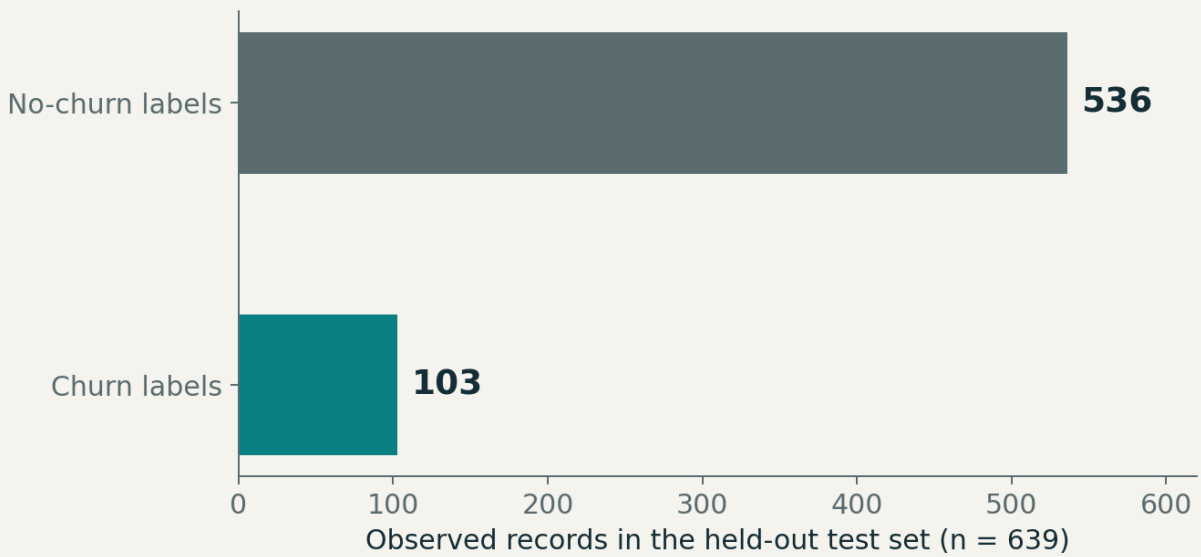
83.9% accurate. Zero churners found.

An all-no-churn prediction looks accurate on this test set.



THE QUESTION

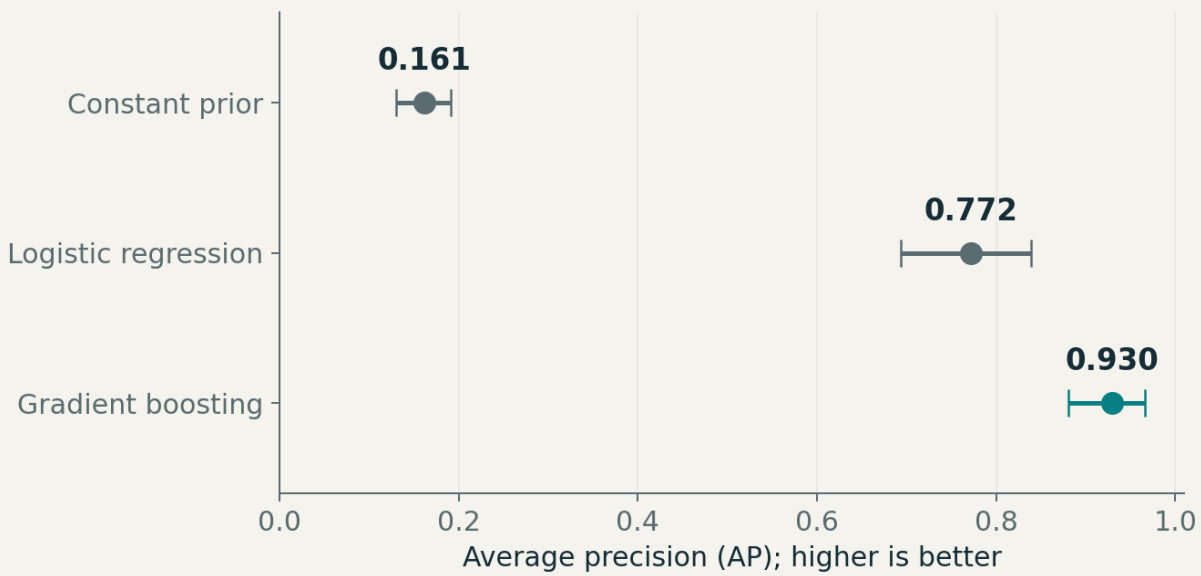
Can a model prioritize a limited review queue and produce defensible risk probabilities?

3,150 observed source records. 639 held out.
A historical Iranian sample, not U.S. carrier evidence.

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Boosting beat the logistic baseline.

Primary models: the same conservative 11-predictor feature set.



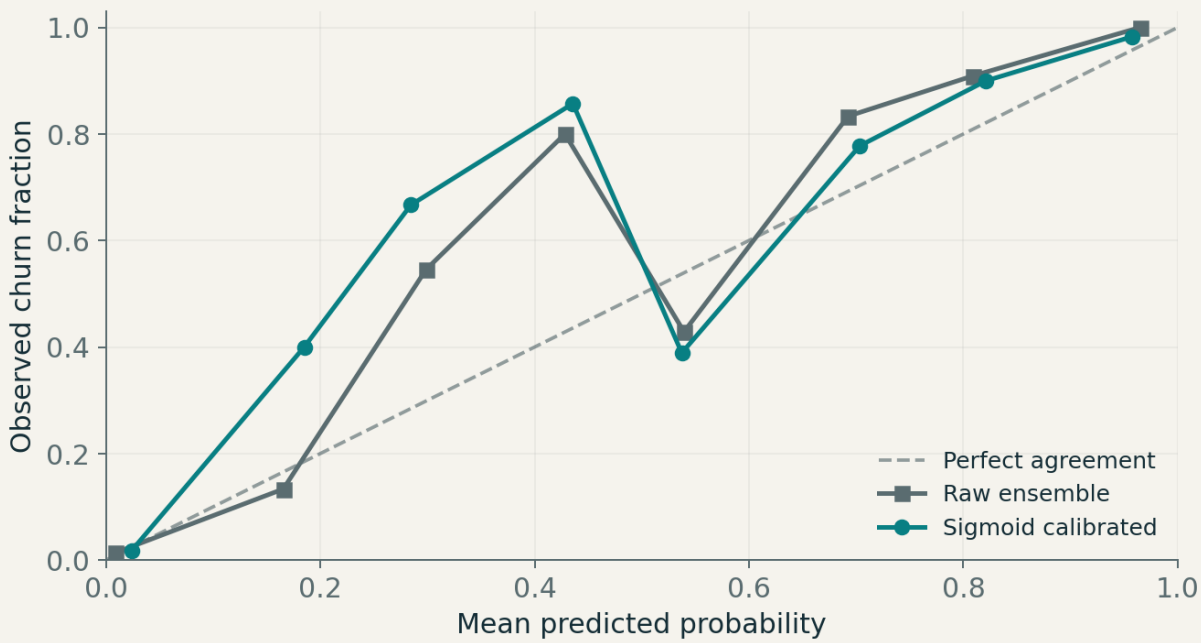
0.930 AP for boosting; 0.772 for logistic.
95% intervals: 0.881–0.966 and 0.694–0.839.

300 repeated rows prompted grouped splits at every layer.
Model choice used development OOF Brier, before test scoring.
Intervals: 500 profile-cluster resamples, conditional on fitted models.

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Calibration didn't improve this result.

Same base-learner ensemble, with and without sigmoid calibration.



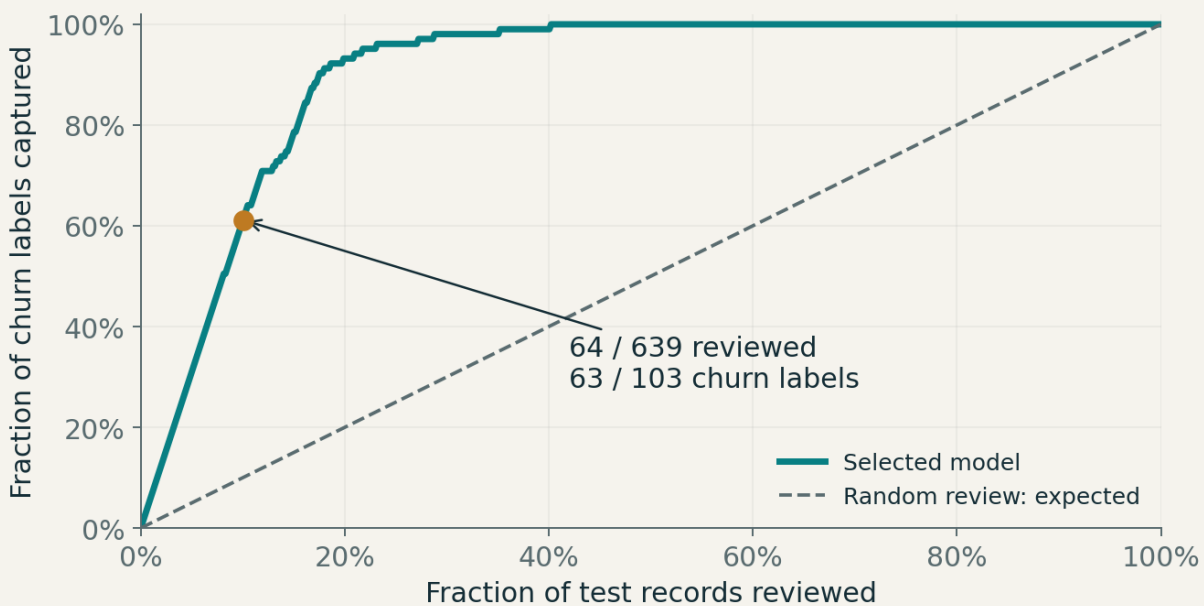
Brier: raw 0.0361 to calibrated 0.0372.
Lower is better. Keep the negative finding.

8 fixed probability bins; some bins have few observations.
Brier measures probability error, not calibration alone.
The prespecified calibrated model remains the primary result.

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64 reviews contained 63 churn labels.

At roughly 10% review capacity in the 639-record test set.



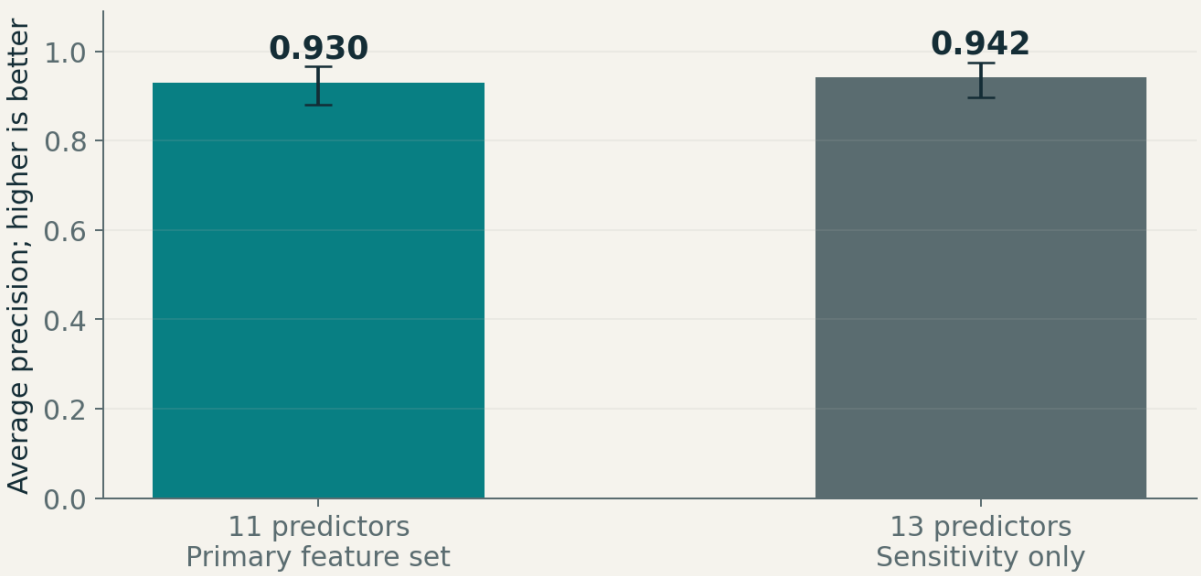
61.2% of the 103 churn labels captured.
Random review would capture about 10% on average.

This is ranking and observed label concentration.
It does not establish prevented churn or offer effectiveness.
An intervention needs separate prospective experimental evidence.

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More fields require an availability check.

Gradient boosting: primary feature set versus source-field sensitivity.



Primary excludes Status and Customer Value.
Their operational construction needs verification.

The source says predictors cover the first 9 months.
Exclusion is a conservative choice, not proof of leakage.
All-field results were ineligible for primary model selection.
Error bars: conditional 95% profile-cluster bootstrap intervals.

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A promising ranking. A bounded conclusion.

The learning: ranking, probability quality, and impact are different.

01 Audit before modelling

Repeated profiles changed the validation design.

02 Nest tuning and calibration

Keep each evaluation fold outside every fit.

03 Make the decision conditional

Validate prospectively before testing an offer.

NEXT: forecast real, dated telecom complaint volumes.

Code, experiment record, predictions and charts on GitHub:
Himalytics/wireless-data-science

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